
SSM Optimization

Weekly Meeting

13/02/26

Application to Falcon Mamba 7B

- Only minor change required to allow for block name matching

```
604 layer_structures = [  
605     # (path to layers, prefix for block names)  
606     (lambda m: m.model.language_model.layers, "model.language_model.layers"), # Mistral3  
607     (lambda m: m.model.layers, "model.layers"), # Standard (Llama, Mistral, etc.)  
608     (lambda m: m.layers, "layers"), # Base models without wrapper  
609     (lambda m: m.backbone.layers, "backbone.layers") # Falcon Mamba model  
610 ]
```

Evaluation

- Used the fast evaluation script (`quant_model_eval.py`) to extract some LLM baselines results.
- Currently encountering some issues in running it with Falcon Mamba 7B, due to some weird dtype inconsistency error:
`RuntimeError: expected mat1 and mat2 to have the same dtype, but got: float != c10::Half`
 - Temporary casting fix, but could be symptom of something not working properly.

Evaluation

* Results slightly different than from SING paper, because to try to solve the Mamba issue I changed the model loading kwargs, and model gets loaded in *bfloat16* instead of *float16*

Model	Quantized	Wikitext2	Memory (GB)
		Perplexity *	
Falcon Mamba 7B		5.203	14.56
Falcon Mamba 7B	 (4 bits)	5.359	4.95
Falcon Mamba 7B	 (3 bits)	5.970	4.22
Qwen3-1.7B		16.714	3.44
Qwen3-1.7B	 (4 bits)	17.137	1.42
Qwen3-1.7B	 (3 bits)	22.394	1.28
Qwen2.5-7B-Instruct		7.461	15.24
Qwen2.5-7B-Instruct	 (4 bits)	7.701	5.86
Qwen2.5-7B-Instruct	 (3 bits)	8.823	5.21
Qwen3-8B		9.727	16.38
Qwen3-8B	 (4 bits)	10.05	6.47
Qwen3-8B	 (3 bits)	21.50	5.73

Benchmarking

Evaluation with Im-eval

Model	Quantized	Lambada OpenAI		MMLU	HellaSwag
		Accuracy (%)	Perplexity	Accuracy (%)	Accuracy (%)
Falcon Mamba 7B	🚫	69.12 ± 0.64	3.203 ± 0.061	59.71 ± 0.39	60.77 ± 0.49
Falcon Mamba 7B	✅ (8 bits)	69.12 ± 0.64	3.201 ± 0.061	59.72 ± 0.39	60.83 ± 0.49
Falcon Mamba 7B	✅ (4 bits)	69.71 ± 0.64	3.193 ± 0.061	59.03 ± 0.39	60.42 ± 0.49
Falcon Mamba 7B	✅ (3 bits)	66.04 ± 0.66	3.681 ± 0.075	54.69 ± 0.40	
Qwen2.5-7B-Instruct	🚫	69.44 ± 0.64	3.687 ± 0.087	71.80 ± 0.36 (74.2 from paper)	61.98 ± 0.48 (80.2 from paper)
Qwen2.5-7B-Instruct	✅ (8 bits)	69.45 ± 0.64	3.699 ± 0.087	71.80 ± 0.36	62.09 ± 0.48
Qwen2.5-7B-Instruct	✅ (4 bits)	67.81 ± 0.65	3.898 ± 0.095	71.07 ± 0.36	
Qwen3-1.7B	🚫	50.51 ± 0.70	12.22 ± 0.508		
Qwen3-1.7B	✅ (4 bits)	47.20 ± 0.70	15.041 ± 0.644		
Qwen3-8B	🚫	64.16 ± 0.67	4.578 ± 0.138		
Qwen3-8B	✅ (4 bits)	62.64 ± 0.67	4.758 ± 0.142		
Qwen3-8B	✅ (3 bits)	26.53 ± 0.62	123 ± 6.908		

Papers

- Falcon Mamba
 - Batch size rampup;
 - Constant LR stages, final LR decay stage leads +6% in average performance;
 - Mamba is more sensitive to learning rates than transformers;
 - Curriculum learning for data mixtures.
- New version of the SING paper
- SSMs quantization papers report results mainly on the Mamba architecture

Next steps

- Analyze the layers' output error between the quantized and floating point model.
- Check also KL divergence.
- Extend the benchmarks table with more models (of similar size) and consider also the Mamba model.
- Application of other methods (?)